

because all other available documents are objectively further removed from the invention (see also T. 570/91, T. 1742/12).

3.5. Further criteria for determining the closest prior art

3.5.1 Defective disclosure

In T. 211/01 the board stated that, apart from the fact that a skilled person would normally not consider an obviously defective disclosure at all, it would in particular be artificial to select a defective disclosure as a starting point for evaluating inventive step when there existed other prior art which was not doubted with regard to its disclosure, but was also directed to the same purpose or effect as the patent in suit. A document which is so obviously defective as to be readily recognised as such by those skilled in the art when trying to reproduce its disclosure cannot be taken as the most promising and appropriate starting point for the assessment of inventive step.

3.5.2 Confidential disclosure in the application

In T. 211/06 the examining division used as a starting point for the assessment of inventive step background art indicated in the application, which according to the applicant comprised confidential information. The board stated that prior art meant publicly available art at the priority date, and not some internal prior art known only within the company employing the inventors. A disclosure in the application as filed which turns out not to be prior art under Art. 54(2) EPC 1973 cannot be used as a starting point for assessing inventive step, nor can subjective "laboratory practice" put forward by the examining division without any objective evidence that this was knowledge available to the public be used as a starting point for assessing inventive step.

3.5.3 Speculative character

In T. 1764/09 the board noted that D1 was no more than a speculative review of what might be potentially feasible in the future. No concrete realisation of a fully adapted lens for a contact or intraocular lens was described in D1. Therefore, for this reason alone, D1, the document selected by the examining division, could not objectively be considered as a realistic starting point or the most promising springboard towards the claimed invention.

In T. 184/10 the board noted that the author of document (14) gave an outlook into future trends, which were of course not yet verified by experimental evidence. In view of the fact that this outlook was based on the data available at the time, the skilled person would not dismiss it as pure speculation. Instead, he would regard it as a serious attempt to interpret the existing state of the art. As a consequence, the content of document (14) qualified as closest prior art.

In T. 725/11 the invention was directed to a pharmaceutical co-formulation in the form of a tablet comprising two active ingredients for HIV therapy. The board did not acknowledge an inventive step over an announcement by the patentee of a clinical trial of that combination therapy in an industry journal article. The patentee argued that this journal

article was not the closest prior art because it was silent on efficacy and did not provide any technical details. The board disagreed and stated that the journal article amounted to a concrete plan to develop a commercially viable product with a usable level of efficacy. Furthermore, the article was a public statement of intent made by the patentee's CEO and its executive vice president of research and development which would not be dismissed by the skilled person as mere speculation.

3.5.4 Old prior art documents

In principle, any document that is state of the art under Art. 54(2) EPC may be a candidate for the closest prior art, without time restriction. The case law acknowledges, however, that in some cases a document may not be a realistic starting point because it either relates to outdated technology, and/or is associated with such well known disadvantages that the skilled person would not even consider trying to improve on it.

In T 69/94 the board pointed out that Art. 54(2) EPC 1973 defined the state of the art as comprising everything made available to the public, clearly without any time restriction. Thus a document which related to an antiquated technology no longer used in industry comprising a teaching disapproved by those skilled in the art at the filing date of the patent in suit could not simply be disregarded as the closest prior art only because of its publication date about 20 years before the filing date of the application documents. In T 1408/04 the board held, however, that a document's age as such was not a ground for excluding it as closest prior art and as a starting point for the assessment of inventive step (see also T 153/97).

In T 334/92 the board held that a document that had been disregarded for more than 20 years by those skilled in the art, had never been used during that period as a basis for further development, was moreover completely silent about the extent of the indicated activity and, finally, did not even mention, let alone discuss, the relevant state of the art, did not represent the closest state of the art and could not therefore be used to define a realistic technical problem. However, in T 964/92, filed as a divisional application to T 334/92, the board held that the same document could be considered as a realistic starting point for the determination of the relevant technical problem.

In T 1000/92 the board did not agree to the selection of document (1) as the closest state of the art because the disadvantages of the process it described, which had been published about 30 years before the priority date of the application, were so evident and well-known that a skilled person would not have tried to improve and develop such an old process (see also T 616/93).

In T 479/00 the board did not regard a 65-year-old document as a realistic starting point for the evaluation of inventive step. It was unrealistic to assume that, without hindsight, somebody of average skill in the art of colouring ceramic articles in 1994 would have had the intention to improve a technique which had not received any attention during the previous 65 years. Furthermore, the teaching of this document, published in 1929, had never been put into practice on a commercial scale.

In T 113/00 the board said that resurrecting very old teaching (in the case in hand 31 years old) with an obvious modification did not make known subject-matter inventive. In T 1397/07 the board added that there was no legal basis in the EPC to preclude a particular state of the art from being taken into account when assessing inventive step, merely because of being published some years earlier than another one.

3.5.5 Improvement of a production process for a known product

Where the invention concerns improving a process to manufacture a known chemical compound, then the closest prior art is confined to documents describing that compound and its manufacture. Comparison with these alone shows whether an improvement has been achieved which can thus be taken into account in formulating the problem the invention sought to solve (T 641/89, T 961/96, T 713/97, T 948/01, T 833/02, T 339/03). In the case of inventions concerning a special process for use with a particular chemical substance having necessarily specific characteristics, determining the closest prior art must involve, above all, considering only those documents which describe a generically corresponding process for using precisely this particular chemical substance with its specific characteristics (T 1285/01, T 354/03, T 1652/08). This accurately and objectively reflects the actual situation in which the skilled person found himself on the priority date of the contested patent (T 793/97).

In T 325/97 the board saw no reason why the above considerations regarding the closest prior art should not also apply to production processes for subject-matter other than a chemical compound and in the case in hand identified the problem to be solved as providing an improved method for the manufacture of a patch for the controlled transdermal delivery of nicotine. In T 373/94 the board also applied the principles and conclusions laid down in T 641/89 where the invention related to the improvement of a manufacturing process for prefilled plastic syringes.

In T 2210/19 the board defined the problem as the provision of a further process for producing a chromium catalysed ethylene copolymer powder. It explained that the skilled person, looking for a further process, would consider variations of the process already known from the closest prior art, taking into account the common general knowledge in the field and the knowledge made available in the prior art. This included variations of any of the parameters of the process that could be expected to be suitable to carry out alternative processes without exercising any inventive activity. As the problem was simply the provision of a further process, no further motivation was needed by the skilled person to perform the modified process.

3.6. Consequences of choosing a certain starting point

In T 570/91 the board emphasised that although a person skilled in the art was completely free in choosing a starting point, he would of course be bound afterwards by that choice. If, for instance, the skilled person preferred and decided to start from a specific compressor piston, he could further develop that piston but at the end of that development the normal result would still be a compressor piston and not an internal combustion engine piston. In T 439/92 it was explained that a conscious choice of starting point, made in the knowledge

of the respective benefits and drawbacks of the various types concerned, not only determined the subject-matter serving as a starting point but also defined the framework for further development, i.e. a further development within this particular type. A change of type during the further development of the consciously chosen type, to another type, which was previously known but had not been chosen, could then only be seen as the result of an ex-post-facto analysis (see also T 1040/93, T 35/95, T 739/95, T 255/03). It is unlikely, and normally not obvious, for the invention type originally chosen to be changed during development (T 817/94, T 749/11, T 535/10).

Following T 439/92, the board stated in T 1228/08 that, although D1 disclosed the use of hierarchical coding, which was a major aspect of the invention, this document did not represent a realistic starting point for an attack on inventive step. A conscious choice of a starting point not only determines the subject-matter serving as a starting point but also defines the framework for further development. Thus, using D1 as a starting point would mean that any further development would be carried out in the context of multicasting: it was unrealistic to suggest that, starting from this disclosure, the skilled person would go outside this framework – indeed take a technical step back – to develop a non-multicasting system.

In T 487/95 the board chose a military protective helmet as the closest prior art, pointing out, however, that this did not mean that documents describing protective helmets of a different kind (such as workers' safety helmets) could not form part of the knowledge of a person skilled in the art. The information contained in the patent which related to a known military protective helmet (D9) represented the **primary** source of information, i.e. the most promising point of departure, from which the skilled person would attempt to arrive at the claimed subject-matter. The other documents could, however, represent important **secondary** sources of information (in this case: workers' safety helmets) from which the skilled practitioner could obtain indications and suggestions with regard to the problem addressed (see also T 149/00).

The invention in T 25/13 related to a device for fastening an actuator and a housing, for use in a motor vehicle. D4, the only document cited by the opponent, which bore the burden of proof for lack of inventive step, described a fastening device for a tumble dryer. The board explained that opponents were essentially free to choose the starting point for assessment, but their choice then had implications for the technical knowledge of the relevant skilled person.

4. Technical problem

Following the selection of the closest prior art and the assessment of the technical effects achieved by the claimed invention in comparison with the closest prior art, the next step of the problem and solution approach is to define the objective technical problem (see in this chapter I.D.4.1 and I.D.4.2).

The "subjective" technical problem as originally presented by the applicant (R. 42(1)(c) EPC) might require reformulation on the basis of objectively more relevant elements originally not taken into account by the applicant or patentee, or where the

combination of features in the claim do not solve this problem over the whole area defined in the claim. This reformulation then defines the "objective" technical problem. Reformulation might lead to the objective technical problem being less ambitious than originally envisaged by the application. See in this chapter I.D.4.4.

See also the Guidelines, G-VII, 5.2 – March 2022 version.

4.1. Determination of the objective technical problem in general

According to the problem and solution approach, **objective criteria** must be used to determine the technical problem (see e.g. T 1/80, OJ 1981, 206; T 24/81, OJ 1983, 133; T 39/93, OJ 1997, 134), i.e. the problem which can be seen to have been actually solved in the light of the closest prior art which may be different from the prior art which was at the disposal of the inventor (T 576/95, T 420/14, T 1148/15). This requires the assessment of the technical effect vis-à-vis the closest state of the art (T 148/05, T 1422/12, T 141/16).

A comparison of the problem indicated in the application with that indicated in a prior document must avoid an unduly abstract approach far removed from the practical thinking of the person skilled in the art (T 5/81, OJ 1982, 249; T 150/89; T 417/94; T 177/98; T 263/99; T 1093/04).

The definition of artificial and technically unrealistic problems is to be avoided (see T 495/91, T 741/91, T 334/92, T 708/96, T 257/98, T 1967/08, T 98/16).

For the purpose of the problem and solution approach, the problem must be a technical one that a skilled person in the particular technical field might be asked to solve at the priority date. (see e.g. G 1/19, OJ 2021, A77; T 385/89; T 641/00, OJ 2003, 352; T 154/04, OJ 2008, 46). In identifying the problem it is not permissible to draw on knowledge acquired only after the date of filing or priority (T 268/89, OJ 1994, 50; T 365/89). However, see in this chapter I.D.4.3.3 "Post-published evidence".

When defining the objective technical problem, an effect cannot be retained if it is not credible that the promised result is attainable throughout the entire range covered by a claim (T 626/90; T 583/93, OJ 1996, 496; T 25/99; T 71/09; T 824/07; T 447/10; T 1837/13, T 340/13). In T 939/92 (OJ 1996,309) the board concluded that a technical effect which justified the selection of the claimed compounds in the case in hand must be one which could be fairly assumed to be produced by substantially all the selected compounds (see also T 489/14 of 22 February 2019, OJ 2019, A86; T 41/16).

In T 161/18 the claimed method was distinguished from the prior art only by an artificial neural network; however, the training of this network was not disclosed in detail. The board was not convinced that the asserted effect was achieved by the claimed method across the whole scope of the claim. This effect therefore could not be taken into account as an improvement over the prior art when assessing inventive step.

In T 1639/07 the board held that the objective technical problem must be derived from physical, chemical etc. effects directly and causally related to the technical features of the